

Project Scheduling

Project scheduling is the process of **planning, organizing and coordinating different tasks of a project** so that the project is completed **on time and within the available budget**.

In agriculture, project scheduling can be used to plan activities such as **planting, irrigation, fertilization and harvesting**.

Objectives of Project Scheduling

The main objectives are:

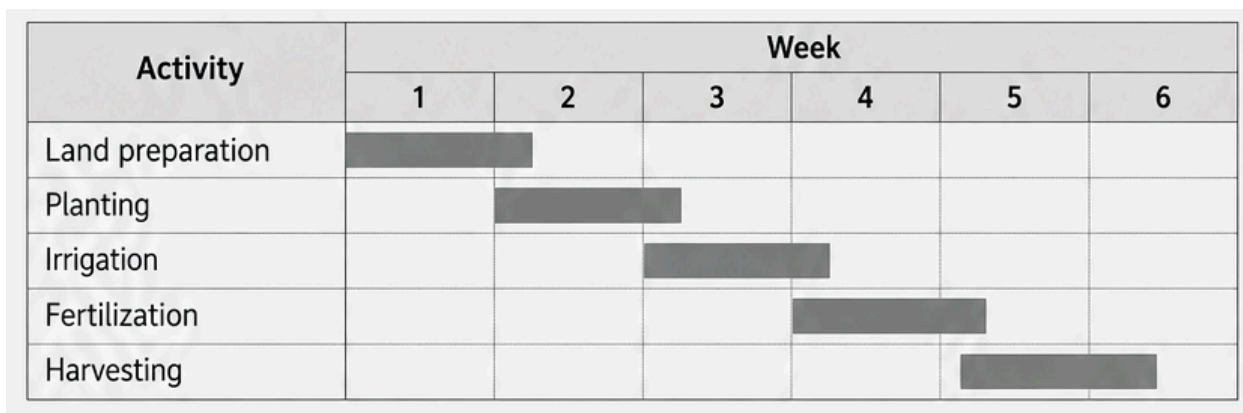
- To plan all activities of the project.
- To decide the time required for each activity.
- To arrange activities in the correct order.
- To identify activities that depend on other activities.
- To complete the project on time.

Main Techniques of Project Scheduling

A. Gantt Chart

A **Gantt chart** is a visual chart that shows the **timeline and duration of different project activities**.

Example: In an agricultural project, a Gantt chart can show the time required for land



B. Network Diagram

A **network diagram** represents the **activities of a project and their relationships**.

It shows:

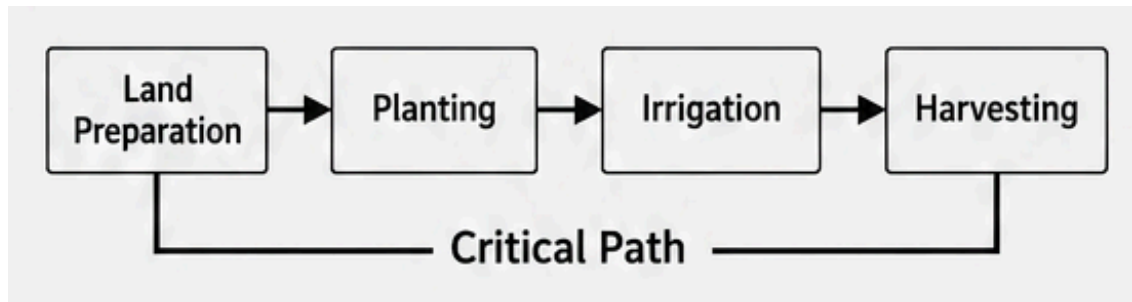
- Which activity comes first.
- Which activity depends on another activity.
- The sequence of activities.
- The total project flow.



C. Critical Path Method (CPM)

CPM identifies the **longest sequence of activities** that determines the total project duration.

Example:

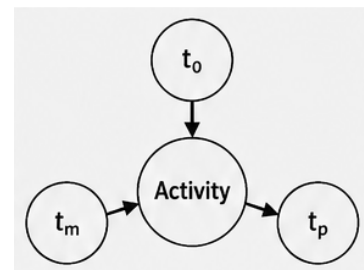


D. PERT

PERT (Program Evaluation and Review Technique) is a **probabilistic network analysis technique**.

PERT considers three time estimates:

1. **Optimistic time** (t_0) – minimum possible time.
2. **Most likely time** (t_m) – normal expected time.
3. **Pessimistic time** (t_p) – maximum possible time.



Applications in Agriculture

Project scheduling is useful in:

Application	Purpose
Planting	Plan the correct time for planting
Irrigation	Schedule water supply
Fertilization	Plan fertilizer application
Harvesting	Decide the harvesting time
Construction	Schedule farm buildings and infrastructure
Research	Organize agricultural research activities

The study material specifically identifies **construction projects, agricultural operations and research projects** as applications of project scheduling.

Advantages

- Helps to complete projects on time.
- Makes the sequence of activities clear.
- Helps in proper use of resources.
- Helps to identify delays.
- Makes project progress easier to monitor.
- Helps managers make better decisions.
- Useful when activity times are uncertain through PERT.

AI and Decision Support Systems

Artificial Intelligence (AI)

Artificial Intelligence (AI) is a technology that enables computers to **learn, reason, solve problems and make decisions** like intelligent systems.

Applications of AI in Agriculture

1. Crop Yield Prediction
2. Pest and Disease Management
3. Precision Agriculture
4. Farm Decision Support

Example of AI and DSS

A farmer wants to decide the **best time to plant a crop**.

The DSS collects:

- Weather data
- Soil information
- Crop information
- Market prices

The system analyzes this information and gives advice about:

- Best planting date
- Fertilizer application
- Irrigation schedule and pest control

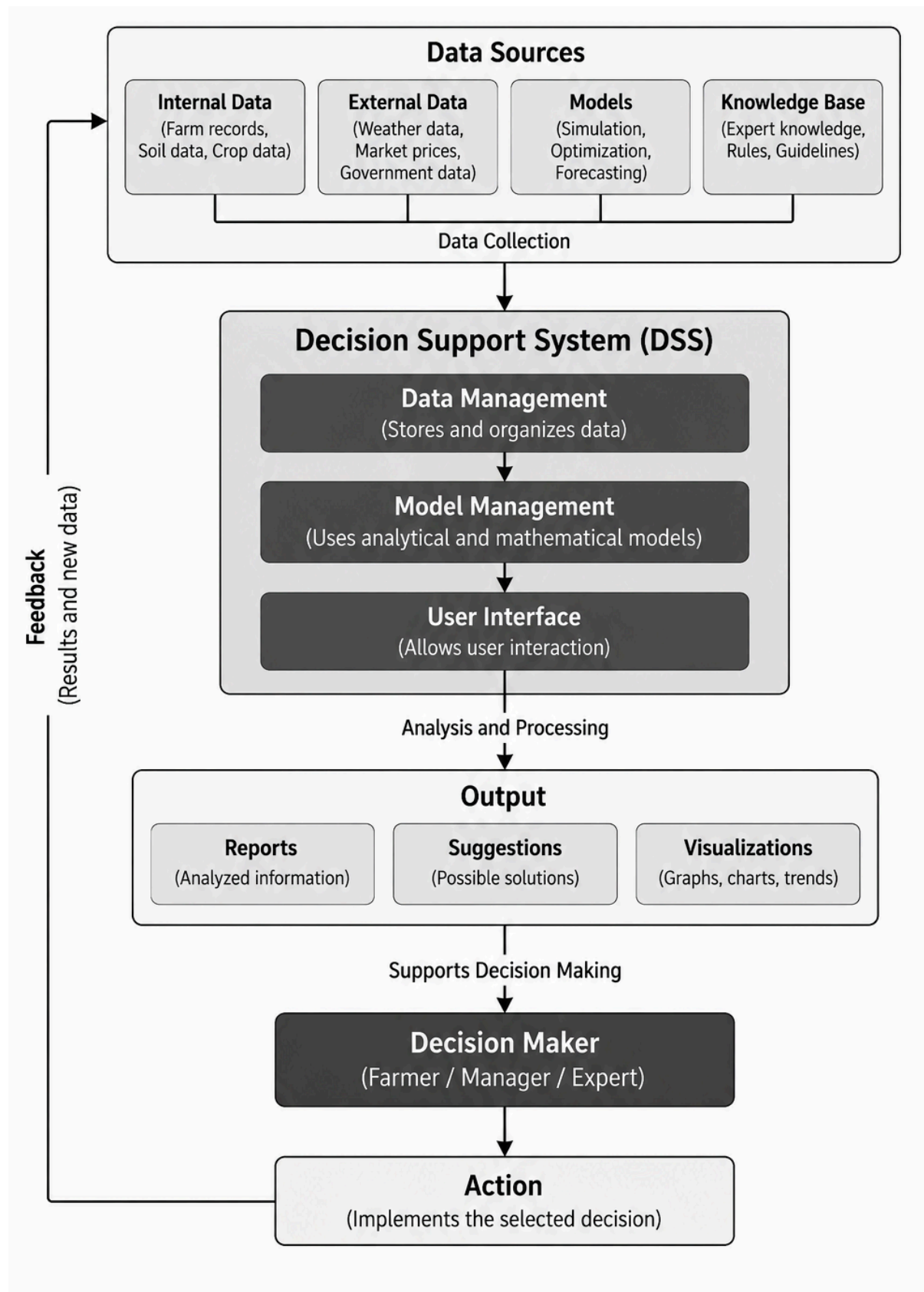
The **farmer makes the final decision** based on the information provided by the DSS.

Decision Support System (DSS)

A **Decision Support System (DSS)** is a **computer-based system that helps people analyze data and make informed decisions**.

A DSS does **not replace the human decision-maker**. It provides information, analysis and suggestions, while the final decision is made by the person.

DSS Workflow



Applications of Decision Support Systems

The study material lists the following major applications:

1. **Diagnostics** – finding crop problems.
2. **Marketing** – helping with agricultural marketing decisions.
3. **Systems operations management** – managing agricultural operations.
4. **Automatic control** – controlling systems automatically.
5. **Strategic planning** – supporting long-term planning.